

Math708 – Foundations of Computational Mathematics I

Fall 2012

Homework 1

1. Write down the Lagrange interpolation polynomial $p_1(x) \in \mathbb{P}_1$ for the function $f : x \rightarrow x^3$ using the points $x_0 = 0, x_1 = a$. Verify Theorem 1.7 in the lecture notes by direct calculation, showing that ξ is unique and has the value $\xi = \frac{1}{3}(x + a)$.
2. Show that for pairwise distinct real numbers $x_0, x_1, \dots, x_n$, the corresponding Lagrange polynomials satisfy:

$$\sum_{i=0}^n L_i(x) \equiv 1;$$
$$\sum_{i=0}^n L_i(0)x_i^j = \begin{cases} 1, & \text{for } j = 0, \\ 0, & \text{for } 1 \leq j \leq n, \\ (-1)^n x_0 x_1 \cdots x_n, & \text{for } j = n + 1. \end{cases}$$

3. Given $n + 1$ support points $\{(x_i, y_i)\}_{i=0}^n$ and let $p \in \mathbb{P}_n$ be the corresponding interpolating polynomial. Prove

$$p(x) = \sum_{i=0}^n \frac{\omega_{n+1}(x)}{(x - x_i)\omega'_{n+1}(x_i)} y_i$$

where $\omega_{n+1}(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$.

4. Show that for given pairwise distinct support nodes $x_0, x_1, \dots, x_n \in \mathbb{R}$ and support ordinates $f_0, f_1, \dots, f_n \in \mathbb{R}$, the corresponding divided difference satisfy

$$f[x_0, \dots, x_n] = \sum_{i=0}^n \left[\frac{f_i}{\prod_{k=0, k \neq i}^n (x_i - x_k)} \right].$$

5. Provide an estimate of

$$\max_{x \in [-1, 1]} |\omega_{n+1}(x)|$$

for $\omega_{n+1}(x) = (x - x_0)(x - x_1) \cdots (x - x_n)$, in the case $n = 1$ and $n = 2$, for a distribution of equally spaced nodes $-1 = x_0 < x_1 < \cdots < x_n = 1$.

6. Construct the polynomial $p_2 \in \mathbb{P}_2$ on the interval $[-1, 1]$ that interpolates the function g defined by $g(x) = \cos x$ at roots of the Chebyshev polynomial T_3 .
7. Suppose that $(x_i, f^{(k)}(x_i))$ are given data with $i = 0, \dots, n$ and $k = 0, \dots, m_i$. Let $N = \sum_{i=0}^n (m_i + 1)$, prove that, if the nodes $\{x_i\}_{i=0}^n$ are distinct, there exists a unique polynomial $H \in \mathbb{P}_{N-1}$, such that

$$H^{(k)}(x_i) = y_i^{(k)}, \quad i = 0, \dots, n, \quad k = 0, \dots, m_i,$$

where $y_i^{(k)} = f^{(k)}(x_i)$.

8. (**Programming Exercise**) Write a program to interpolate the function

$$f(x) = \frac{1}{25x^2 + 1}, \quad x \in [-1, 1],$$

with a polynomial of degree $\leq n$ using the Newton's interpolation formula

- at the equidistant points $x_j = -1 + 2j/n$, $j = 0, 1, \dots, n$.
- at the zeros $t_k^{(n+1)}$, $k = 1, 2, \dots, n+1$ of the $(n+1)$ -st Chebyshev polynomial T_{n+1} .

Plot the respective graphs for $n = 5, 10$. Also submit your codes in both paper and email. You could use Language such as C/C++, Fortran, Matlab or Maple.